



EduPath

OFFICIAL PRODUCT DOCUMENTATION • V1.0

User Manual & Operational Guide

Comprehensive operational handbook for EduPath: An Autonomous AI-Powered Career Upskilling & Adaptive Learning Platform.

"Your skills today. Your career tomorrow. One adaptive path."

Prepared For: Students, Engineers, Career Switchers & Hackathon Evaluators

Target Systems: Windows, macOS, Linux • Next.js 14 App Router

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Note for Hackathon Judges & Reviewers: EduPath is not a static UI mockup or generic LLM wrapper. It is a deterministic, event-driven autonomous learning agent engine with real state persistence, dynamic gap calculations, and adaptive schedule recovery loops.

1. Executive Summary & System Purpose

In modern software engineering and technology fields, learners frequently know the target career role they aspire to achieve (e.g., *Full Stack Developer*, *Data Scientist*, *AI/ML Engineer*), but face three systemic barriers:

1. **Invisible Baseline:** Learners struggle to objectively map what skills they already possess versus what the industry requires.
2. **Generic Curricula:** Existing courses offer one-size-fits-all 40-hour video playlists that do not adapt when a learner struggles or excels.
3. **Learning Debt Accumulation:** When a learner fails to grasp a prerequisite (e.g., SQL JOINS), traditional platforms keep marching forward, causing cascading frustration and abandonment.

EduPath solves this by functioning as an **Autonomous Career Learning Operating System**. It ingests the learner's profile, calculates quantified skill gaps against industry benchmarks, synthesizes a personalized weekly roadmap with curated resources, and continually monitors practice telemetry. When a learning bottleneck is detected, it autonomously rewrites the roadmap, inserts recovery modules, and explains every decision via human-readable explainability receipts.

Key Platform Pillars

DYNAMIC Profile Customization

Enter your own name, select any target role, and type your actual skills to generate a 100% custom roadmap and gap analysis.

CLOSED-LOOP Adaptive Replanning

Diagnostic quiz failures trigger real-time graph rebalancing, schedule pushouts, and recovery module injections.

ZERO BLACK BOX Explainability

Every schedule change produces an audit receipt showing trigger cause, confidence score, trade-offs, and affected milestones.

CURATED High-Yield Resources

Directly links official documentation (PostgreSQL, FastAPI, Scikit-Learn) rather than generic fluff.

2. Quick Start Guide

EduPath is packaged for immediate local execution on Windows with zero external service dependencies required for core functionality.

Launching the Platform in 1 Click

1. Navigate to your Windows Desktop and locate the shortcut file: `Start_EduPath.bat`.
2. Double-click the batch file. A terminal window will open, verifying dependencies and starting the Next.js development server.
3. Open your preferred web browser (Google Chrome, Microsoft Edge, Brave) and navigate to:
<http://localhost:3000>

Live Status Check: If the server is active, accessing `http://localhost:3000` will render either the high-impact Landing Page or the Career OS Shell directly.

Command Line Alternative

If you prefer running commands in PowerShell or Command Prompt:

```
cd c:\Users\de11\Desktop\edupath  
npm run dev
```

3. System Architecture: The 9 Live AI Agents

EduPath is engineered around a collaborative **Multi-Agent Architecture** where each specialized agent executes distinct responsibilities across the learning lifecycle:

Agent Name	Primary Responsibility	Inputs	Outputs Produced
1. Profile Agent	Extracts structured skills and verified evidence from intake and resumes.	User input, resume text, project summaries	Normalized skill list with verified quotes and proficiencies
2. Skill Intelligence Agent	Normalizes terminology and benchmarks proficiencies against industry roles.	Extracted skills, role taxonomy	Scored proficiencies (0.0 to 1.0) and confidence intervals
3. Gap Agent	Calculates exact competency deltas and classifies urgency (Critical / High / Medium).	Learner skill matrix, target role benchmark	Prioritized skill gap inventory with rationale
4. Roadmap Planner Agent	Generates a sequential Directed Acyclic Graph (DAG) of weekly milestones and tasks.	Skill gap list, time budget (min/day)	Chronological roadmap with prerequisites
5. Resource Agent	Matches high-yield documentation, tutorials, and interactive exercises to gaps.	Skill objectives, preferred modality	Curated resource links with source attribution
6. Practice Agent	Generates diagnostic MCQs and coding challenges with sub-concept tracking.	Target skill, difficulty level	Diagnostic questions, answer keys, explanations
7. Progress Agent	Tracks learner velocity, completed missions, daily streak, and awarded XP.	Task completions, diagnostic results	Telemetry metrics, streak updates, readiness %
8. Adaptive Planner Agent	Detects repeated diagnostic struggles and autonomously rewrites the schedule.	Diagnostic failure signals	Recovery modules, date shifts, decision receipts
9. AI Mentor Agent	Provides conversational coaching, answers learner questions, and explains changes.	Learner queries, context state, decisions	Contextual chat responses with actionable recommendations

Auditability: Click the **AGENTS (9)** button in the top navigation header at any time to open the **Live Agent Activity Drawer**, showing real-time execution logs, latency benchmarks, and payload summaries for all 9 agents.

4. Core Application Views & Features

4.1 " ⚡ Try YOUR Profile" (Custom Path Engine)

Located conspicuously in the top-right header, this modal allows any user to transition away from the pre-scripted demo and generate a 100% personalized learning plan in seconds:

- **Your Name:** Personalizes all mentor greetings and milestone headers (e.g., *Pranav* or *Rahul*).
- **Target Career Role:** Select from *Full Stack Developer*, *Data Scientist*, or *AI / ML Engineer*.
- **Skills You Already Know:** Enter comma-separated skills (e.g., `Python, SQL, Git, HTML`). Quick presets are provided for one-click testing.
- **Daily Commitment:** Set available daily study time (15, 30, or 60 min/day).

Upon submission, the engine dynamically recalculates the learner's readiness percentage, identifies exact gaps against role benchmarks, generates a custom weekly roadmap, logs an agent execution event, and triggers a congratulatory confetti animation.

4.2 Career Overview Dashboard

The central command hub displaying high-level performance indicators:

- **Top KPI Cards:** Career Readiness %, Active Skill Gaps, Completed Tasks, and Learning Debt counter.
- **Target Role Banner:** Displays learner name, target trajectory, daily commitment, and quick links.
- **Active Milestone Mission:** The immediate next actionable task with direct links to official documentation.
- **Quick Diagnostic Teaser:** Fast check to test foundational knowledge.

4.3 Interactive Skills Map

Provides transparency into how the AI evaluates the learner's abilities:

- **Proficiency Breakdown:** Visual bars comparing learner mastery against industry benchmark thresholds (e.g., 68% vs 80% benchmark).
- **Evidence Extraction Citations:** Quotes demonstrating where and how the skill was verified (e.g., "Extracted from GitHub repo: `express-microservice`").
- **Confidence Rating:** AI confidence metric (e.g., 90% confidence) based on evidence density.

4.4 Skill Gap Analysis

Categorizes skill deficits by urgency:

- **CRITICAL GAPS:** Blocker competencies required for entry-level hiring (e.g., `SQL JOINS` for Backend).
- **HIGH & MEDIUM GAPS:** Important secondary skills (e.g., `Docker`, `Redis caching`).
- **Rationale:** Plain-language explanations detailing *why* the specific skill is mandatory for the role.

4.5 Adaptive Directed Acyclic Graph (DAG) Roadmap

The chronological learning schedule structured into sequential weekly milestones:

- **Sequential Milestones:** Week 1 (Active), Week 2 (Upcoming), Week 3 (Future).
- **Task Checklists:** Interactive checkboxes allowing learners to complete tasks. Checking off tasks increases XP and plays celebratory feedback.
- **Curated Resource Cards:** Direct links to vetted industry documentation (PostgreSQL official docs, Express.js guides, LangChain docs) with estimated completion time.
- **Recovery Module Indicators:** Special amber cards injected when an adaptive intervention occurs.

4.6 Practice Lab, Diagnostic Telemetry & Quizzes

The diagnostic engine that validates whether a learner actually understands what they study:

- **Sub-Concept Tagging:** Questions are tagged with granular concepts (e.g., *SQL* → *Multi-Table JOINS*).
- **Instant Feedback:** Selecting an answer provides instant correctness feedback, detailed technical explanations, and hints.
- **Struggle Telemetry:** Answering multiple questions incorrectly on a concept automatically alerts the **Adaptive Planner Agent**.

4.7 Bottleneck Detection & Autonomous Replanning

The crown jewel of EduPath's intelligence. When the system detects repeated errors:

1. A persistent top amber alert bar activates: *"Learning Bottleneck Detected: Repeated difficulty with SQL JOINS."*
2. The **Adaptive Planner Agent** modifies the roadmap graph in real-time.
3. A focused **45-minute SQL Recovery Module** is inserted into Week 1.
4. Downstream complex milestones (e.g., Week 2 Backend APIs) are automatically shifted by 2 days to protect learner cognitive load.
5. The **"Why Did My Plan Change?"** modal automatically surfaces to explain the rationale.

4.8 "Why Plan Changed" Explainability Receipts

Prevents the learner from feeling confused or disoriented by automated changes:

- **Trigger Cause:** Specific failure telemetry (e.g., "2 consecutive diagnostic failures on SQL JOIN queries").
- **Action Taken:** Specific modifications made (e.g., "+45 min recovery module injected; Week 2 postponed 2 days").
- **AI Confidence Score:** Evaluated certainty of the intervention (e.g., 94% confidence).
- **Trade-Off Disclosure:** Clearly states that graduation date moves out by 2 days to ensure foundational mastery.

4.9 Context-Aware AI Mentor Assistant

Accessible via the "AI Mentor" tab. The mentor is fully aware of:

- The learner's current target role, name, and readiness score.
- Whether an active bottleneck exists.
- The latest adaptive replanning decision.
- Learners can ask for code examples, concept explanations, or career advice.

5. Live Demonstration Scenarios (For Judges & Evaluators)

When presenting EduPath in a hackathon demo or live review, execute these three scripted sequences for maximum impact:

Scenario A: The Personalized Custom Path (2 Minutes)

1. Open `http://localhost:3000`.
2. Click the top-right button: " ⚡ Try YOUR Profile".
3. Type your name (e.g. *Pranav*), choose **AI / ML Engineer**, and click the preset chip **"AI/ML Enthusiast"**.
4. Click **"Analyze My Gaps & Build My Path"**.
5. **Highlight to Audience:** Point out how the entire platform transformed into an AI/ML engineering curriculum with Vector Databases (pgvector), RAG Pipelines, and FastAPI serving tasks.

Scenario B: The Adaptive Replanning Loop (2 Minutes)

1. Look at the bottom floating toolbar (**Demo Control Bar**).
2. Click the red button: " ⚡ Simulate Diagnostic Struggle".
3. **Highlight to Audience:** Point out the top bottleneck banner, the automatic popup of the **"Why Plan Changed"** receipt, and how Week 1 roadmap now includes a focused 45-minute Recovery Module.
4. Open the **AI Mentor** tab to show the mentor coaching the learner on relational joins.

Scenario C: Rebalancing Learning Debt (1 Minute)

1. In the bottom Demo Control Bar, click " 🔄 Rebalance Learning Debt".
2. Show the green confirmation banner explaining how overdue tasks were redistributed without overwhelming the daily schedule.

6. Technical Specifications & Security Posture

Layer	Technologies Used	Key Features & Architecture
Frontend	Next.js 14 (App Router), React 18, TypeScript, Tailwind CSS	Server & Client Components, Responsive layout, Zero layout shifts
Visuals & Motion	Lucide Icons, Recharts, Framer Motion, Canvas Confetti	Radar charts, proficiency progress meters, celebratory micro-interactions
Agent Engine	Modular TypeScript Agent Pipeline, DAG Rebalancing Engine	Deterministic state machines, telemetry logging, confidence scoring
Database Schema	Supabase PostgreSQL with <code>pgvector</code> extension	Row Level Security (RLS) policies, HNSW vector cosine distance indexing
Security	OWASP Hardened HTTP Headers, Zod Payload Validation	<code>X-Content-Type-Options: nosniff</code> , <code>X-Frame-Options: SAMEORIGIN</code> , disabled <code>X-Powered-By</code>

7. Troubleshooting & Frequently Asked Questions

Q: How do I reset the platform back to the initial demo state?

Click the **"Reset Demo State"** button in the bottom floating Demo Control Bar. This instantly restores the default Alex Rivera profile, clears all simulated bottlenecks, and resets the roadmap.

Q: The terminal says "Port 3000 is already in use". How do I resolve this?

Next.js will automatically fall back to port 3001, or you can terminate existing Node processes in PowerShell by running:

```
Get-Process -Name node | Stop-Process -Force
```

Q: Can I run EduPath without an internet connection?

Yes! All core agent logic, profile generators, diagnostic quizzes, and state rebalancers run locally inside the Next.js runtime. External resource links (e.g. to PostgreSQL documentation) simply open official docs in a new tab when clicked.

Q: Where are the database schema files stored?

The complete hardened PostgreSQL schema with Row Level Security (RLS) policies and pgvector embedding definitions is located at:

```
c:\Users\dell\Desktop\edupath\docs\schema.sql .
```